

SIH 2026 — Problem Statement SIH26143

AIS, Vessel Tracking, GIS & Oil-Spill Backtracking

Research Brief — Vessel Attribution for Satellite-Detected Oil Spills

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Prepared for the internal hackathon — 26 August 2026

0. Scope & Deliverable

This document is the research brief for Person 3's ownership area on Team [Odoos/SIH26143] project: AIS data understanding, usable data sources, vessel-trajectory reconstruction, spill-to-vessel matching, and — following the recent scope update — full ownership of spill-origin backtracking (previously overlapping with Person 1's brief; that overlap has been removed). Everything here is scoped to what is realistically buildable tonight and tomorrow, with a clear separation between what gets coded and what is reported as a documented limitation.

Hard deadline: internal hackathon on 26 August 2026. The target deliverable is a ranked shortlist of candidate vessels for a given spill location and detection time — not a full maritime intelligence platform.

1. Understanding AIS

AIS (Automatic Identification System) is a VHF-based tracking system mandated under SOLAS for vessels above 300 GT, all passenger ships, and fishing vessels above 15 m. Class A units (mandatory) broadcast position roughly every 2–10 seconds while under way and every 3 minutes at anchor; Class B units (smaller/voluntary craft) report less frequently. Each position report carries the fields below.

Field	What it is	Useful for spill attribution?
MMSI	9-digit Maritime Mobile Service Identity; primary dynamic vessel identifier broadcast in every AIS message.	Essential — the join key used to group raw messages into per-vessel trajectories.
IMO number	7-digit identifier permanently tied to the hull for the vessel's life (unlike MMSI, which can change).	Essential for identity persistence — used to cross-reference vessel registries, ownership, and flag state once a candidate is shortlisted.
Latitude / Longitude	Reported position (WGS84) at the message timestamp.	Essential — the core spatial signal for distance-to-spill and trajectory-intersection checks.
Timestamp	UTC time of the position report.	Essential — the core temporal signal; used to compute time-since-passed-spill-site and to reconstruct ordered trajectories.
Speed over Ground (SOG)	Vessel speed in knots as measured by GPS.	High value — near-zero SOG near a spill site (loitering/drift) is a strong evidence signal; also needed for

Field	What it is	Useful for spill attribution?
		interpolation and impossible-jump filtering.
Course over Ground (COG)	Actual direction of travel over the ground (affected by drift/current), in degrees.	High value — used to check whether a vessel's path plausibly intersects or aligns with the spill polygon.
Heading	Direction the vessel's bow is pointing (compass/gyro reading), independent of drift.	Moderate — COG vs. heading divergence can indicate strong cross-currents or unusual manoeuvring near the spill; secondary signal to COG.
Navigational status	Self-reported code, e.g. "under way using engine", "at anchor", "engaged in fishing", "not under command".	Moderate-high — status changes near the spill window (e.g., unexpected stop) are useful corroborating evidence, though self-reported and sometimes stale.
Vessel type	Category code, e.g. tanker, cargo, fishing, tug, passenger.	High value for prioritisation — tankers and cargo vessels carrying bunker fuel are the realistic suspect pool; fishing/passenger vessels are usually deprioritised.
Rate of turn (ROT)	Rate at which heading is changing.	Low-moderate — occasionally useful to flag erratic manoeuvring, not core to the MVP.
Draught	Vessel's current draught (loaded depth).	Low — can hint at loaded vs ballast state; not prioritised for the hackathon timeframe.

Table 1. AIS fields and their relevance to oil-spill vessel attribution.

1.1 Which fields actually matter for this project

For the hackathon MVP, the pipeline can run on a reduced field set without losing meaningful accuracy:

- Core join/identity: MMSI (primary key), IMO (secondary, for post-shortlist verification).
- Core spatio-temporal: latitude, longitude, timestamp — required for every stage (cleaning, trajectory reconstruction, matching, backtracking).
- Core motion: SOG, COG — required for interpolation, impossible-jump filtering, and the motion-evidence score.
- Context/prioritisation: vessel type, navigational status — used to rank and explain candidates, not to filter them out entirely (a false negative from over-filtering is worse than a noisy candidate list).
- Heading and ROT are kept as secondary/explanatory signals — nice to have if time allows, not blocking.

2. Usable AIS Datasets / Sources

Three sources, each verified to actually exist and be accessible as of August 2026 (checked directly against each provider's own documentation).

Source 1 (primary — live/near-real-time feed): AISstream.io	
URL	https://aisstream.io (docs: aisstream.io/documentation)

Geographic coverage	Global, including the Indian Ocean, Arabian Sea and Bay of Bengal — coverage is via a network of AIS receiving stations plus terrestrial/roaming feeds, so density is higher near the coast (Mumbai, Kochi, Chennai, Vizag, Kolkata approaches) and thinner far offshore. Filterable to any bounding box, so we can subscribe to just the Indian coastline / Arabian Sea / Bay of Bengal.
Time period	Live stream only (no historical backfill) — we capture and store data ourselves from the moment we connect.
Fields	MMSI, latitude, longitude, SOG, COG, true heading, navigational status, rate of turn, timestamp, plus static data (IMO, ship name, callsign, ship type, dimensions, destination, ETA) via separate message types.
Format	JSON over a WebSocket connection (wss://stream.aistream.io/v0/stream).
Size	Streaming, not a fixed download — volume depends on bounding-box size and how long we listen (world-wide feed is roughly a few hundred messages/second at peak).
License	Free tier for real-time streaming; sign-up required for an API key. No cost for the volumes this hackathon needs.
Download / API	WebSocket API — subscribe with API key + bounding box(es) + optional MMSI/message-type filters; official Python/JS/Go examples exist (github.com/aistream).
Restrictions	Verified: this is a genuinely free, self-serve, sign-up API — no need to run a physical receiver (unlike AISHub, see below). Terrestrial-feed dependent, so mid-ocean gaps are possible; treat as "best-effort near-real-time", not guaranteed complete coverage.
Can we realistically use it?	Yes — this is our recommended primary source for anything demo'd live. Requires starting a listener script early (tonight) so we accumulate enough Indian-Ocean traffic to have a non-trivial vessel pool by demo time.

Source 2 (best-fit past-Indian-Ocean-incident validation): NOAA / Global Fishing Watch AIS-derived datasets	
URL	https://globalfishingwatch.org/our-apis/ and https://globalfishingwatch.org/data-download/
Geographic coverage	Global, including the full Indian Ocean, Arabian Sea, and Bay of Bengal — GFW explicitly supports RFMO regions and custom lat/lon bounding-box queries, so an India/Arabian-Sea/Bay-of-Bengal box can be pulled directly.
Time period	2012–present for AIS-based vessel presence and fishing-effort data; near-real-time (72-hour delay) available for heatmap-style layers, ~3-month delay for vessel-level data via the API.
Fields	Vessel presence (lat/lon grid cell, hour, vessel ID, flag, gear/vessel type), apparent fishing effort/hours; the Events API separately exposes encounters, loitering, port visits and AIS-disabling ("dark") events — the loitering/AIS-gap events are particularly relevant to spill investigation.
Format	JSON via REST API and Python/R client libraries (gfw-api-python-client , gfwr); bulk CSV/JSON download also available from the Data Download Portal.
Size	Varies by query; bulk global datasets run into GBs, but a bounding-box + date-range query for the Arabian Sea / Bay of Bengal returns a manageable subset.
License	Free, but API access is for non-commercial use only (fine for a hackathon/academic prototype); data is otherwise openly published by GFW (a 501(c)(3) non-profit).
Download / API	REST API with a free registered access token; also downloadable in bulk without an account for the static public datasets (fishing-effort CSVs, README-documented).

Restrictions	Verified — real, actively maintained, well-documented (api-doc.globalfishingwatch.org). It is fishing-vessel-biased by design (started as a fisheries-monitoring project), so tanker/cargo coverage is present but not its main strength — best used as a secondary/cross-check source and for realistic event types (loitering, AIS gaps) rather than as the main tanker-tracking feed.
Can we realistically use it?	Yes, as a secondary/validation source — good for demonstrating the AIS-gap / loitering event concept and for testing our pipeline against a real, stable, well-documented API before relying on the live AISstream.io feed.

Source 3 (methodology / pipeline-testing dataset — not Indian Ocean, used for offline development): Danish Maritime Authority historical AIS archive	
URL	https://www.dma.dk/safety-at-sea/navigational-information/download-data (raw files: web.ais.dk/aisdata/)
Geographic coverage	Danish waters / Baltic and North Sea approaches — NOT the Indian Ocean. Included deliberately as our third source because it is a real, large, free, well-documented, and widely used academic dataset for prototyping the trajectory-cleaning and reconstruction pipeline before pointing it at live Indian-Ocean data.
Time period	Daily CSV files from 2006 to present, continuously published.
Fields	Timestamp, Type of mobile, MMSI, Latitude, Longitude, Navigational status, ROT, SOG, COG, Heading, IMO, Callsign, Name, Ship type, Cargo type, Width, Length, Type of position fixing device, Draught, Destination, ETA, Data source type — i.e., essentially the full field set we listed in Section 1, in one flat CSV.
Format	CSV, one file per day (e.g. aisd-2023-02-15.csv).
Size	≈1.5–2.6 GB per single day (~10 million rows/day); we would only download one day (or a few hours, pre-filtered) for pipeline testing, not the full archive.
License	Published under Danish open-data / PSI reuse terms — free, unrestricted use, no application needed for the current rolling archive.
Download / API	Direct HTTPS file download, no login, no API key.
Restrictions	Verified — real, currently published, exact CSV schema confirmed. Wrong geography for the actual India-focused demo, so it is used only to build/debug the cleaning → trajectory → matching pipeline offline, then re-pointed at Source 1 / Source 2 data for the real Arabian Sea / Bay of Bengal scenario.
Can we realistically use it?	Yes, but for development only — realistic volume and a clean, well-known schema make it the fastest way to get the trajectory-reconstruction code working tonight, without waiting on live-stream accumulation.

2.1 Practical recommendation for tonight

1. Start an AISstream.io listener immediately, subscribed to a bounding box covering the Indian coastline, Arabian Sea and Bay of Bengal, writing raw JSON messages to disk/DB continuously — every extra hour of runtime is more real Indian-Ocean vessel data for the demo.
2. In parallel, build and debug the cleaning/trajectory pipeline against a single day of the Danish CSV data (Source 3) — it is instantly available, has an identical schema shape, and de-risks pipeline bugs before the live feed has accumulated enough data.
3. Use Global Fishing Watch (Source 2) as a fallback / cross-check and as the source for a realistic loitering or AIS-gap event to showcase, since it is pre-computed and well documented.

3. Vessel Trajectory Reconstruction

Raw AIS is a stream of independent position reports, not trajectories — reconstruction means turning per-message noise into ordered, per-vessel paths that are safe to reason about geometrically. The pipeline below is the standard approach used across the AIS-analytics literature (and mirrors what MovingPandas expects as input).

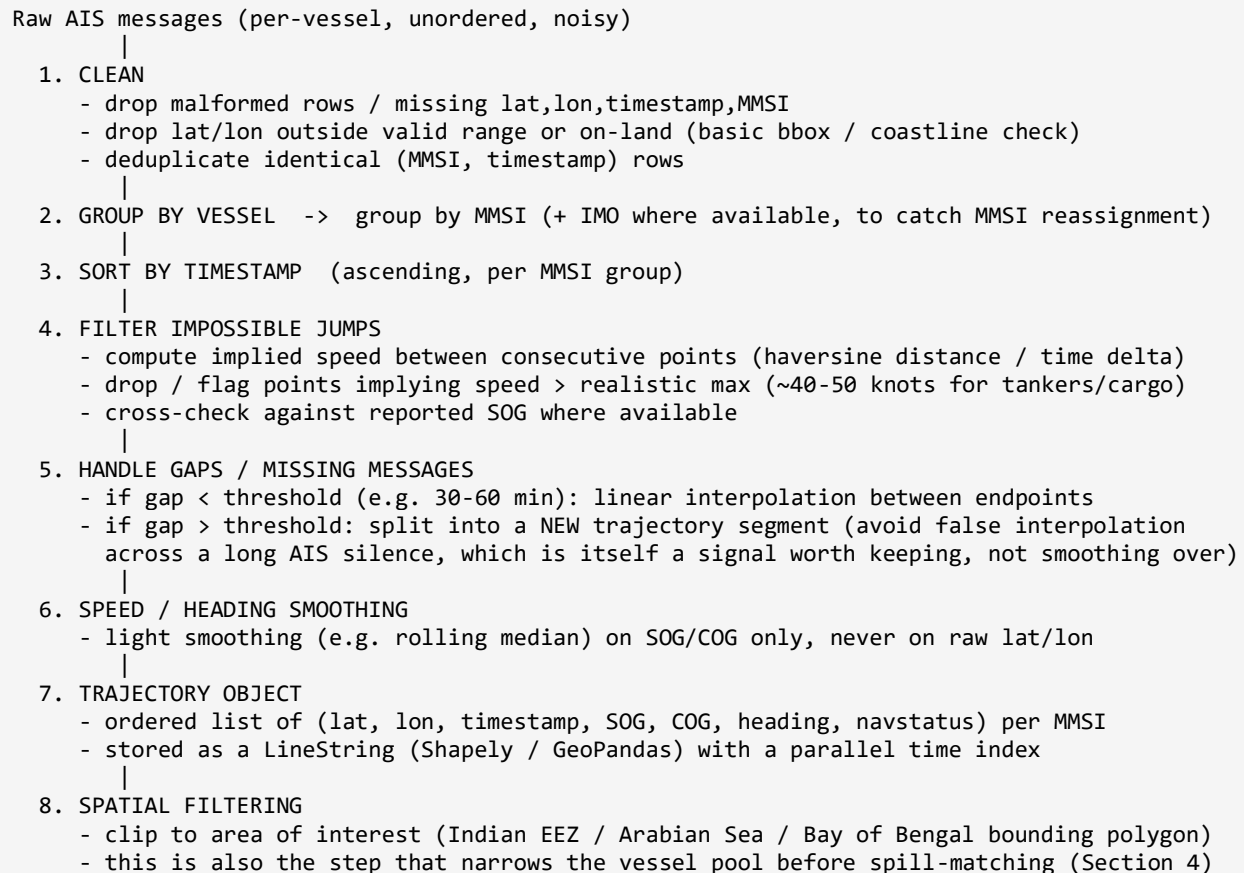


Figure 1. Trajectory reconstruction pipeline, raw AIS to clipped per-vessel trajectories.

3.1 Key decisions and why

- Gap threshold splits, not always interpolates: a long AIS silence near a spill is itself evidence (possible deliberate AIS shutoff), so the pipeline preserves that gap as a boundary between two trajectory segments rather than smoothing straight through it.
- Impossible-jump filtering uses both implied speed (from consecutive points) and reported SOG as a cross-check — a message with a huge positional jump AND a normal reported SOG is more likely a GPS/transmission glitch than real motion.
- MMSI is the primary group key, but MMSI can occasionally be reused/reassigned between vessels, so IMO (when present in static messages) is kept as a secondary identity check for any vessel that becomes a serious candidate.
- Smoothing is applied only to derived motion fields (SOG/COG), never to raw lat/lon — the position track itself should stay faithful to what was actually reported.

4. Matching Vessels to a Spill

Given a spill location (lat, lon) and detection time T, the goal is a ranked, scored shortlist — not a single "guilty" vessel. The output format is designed to be consumed directly by Zorv's attribution model as structured evidence per candidate, rather than a final verdict.

Evidence type	Signal	How it's computed	Contribution to score
Spatial	Distance from spill	Haversine distance from each candidate trajectory point to spill centroid at closest-in-time point.	Higher weight the closer the vessel was to the spill location.
Spatial	Trajectory intersection / buffer	Buffer the spill polygon by an uncertainty margin (e.g. a few nautical miles) and test intersection with each vessel's trajectory LineString (Shapely .intersects / .buffer).	Binary/boosted signal — vessels whose path physically crosses the spill buffer rank above those merely "nearby".
Temporal	Presence before detection	Filter to vessels with at least one AIS point within a lookback window (e.g. 6–24 h) before detection time T.	Gate/filter — vessels with no presence in the window are excluded, not just down-ranked.
Temporal	Time since passed spill site	For vessels that intersected the spill buffer, compute elapsed time between that crossing and T.	Closer in time to T scores higher; an exponential or linear decay weight works for the hackathon timeframe.
Temporal	Estimated spill time window	If spill imagery/appearance gives a coarse time range, restrict candidate presence to that window instead of only T.	Narrows the candidate pool and reduces false positives from unrelated passing traffic.
Motion	Heading / COG alignment	Compare vessel COG at last position before T to the bearing from that position to the spill centroid.	Small angular difference = higher score (vessel was heading toward/through the spill area).
Motion	Speed anomaly	Flag unusually low SOG (loitering/drift) or a sudden SOG drop near the spill site/time.	Loitering near the spill site close to T is a strong positive signal.
Motion	Vessel type prior	Tanker / cargo / bulk carrier types weighted higher than fishing/passenger for oil-spill scenarios (adjustable prior, not a hard filter).	Multiplicative prior on the combined spatial+temporal+motion score.

Table 2. Spatial, temporal, and motion evidence contributing to the candidate-vessel score.

4.1 Matching pipeline, step by step

1. Filter candidate pool spatially: keep vessels whose trajectory has any point within a generous radius (e.g. 20–50 nm, tunable) of the spill location, within the lookback window.
2. Filter candidate pool temporally: keep only vessels present in the area within the lookback window before T (this is a hard gate, not a score — no presence, no candidacy).
3. Score remaining candidates on spatial proximity, trajectory-intersection, time-since-passage, heading alignment, speed anomaly, and vessel-type prior.
4. Combine into a single weighted score per vessel (simple weighted sum is sufficient for the hackathon; weights are tunable constants, not learned, given the timeframe).

- Rank and output the top-N candidates with their individual evidence components intact — not just the final score — so Zorv's model can re-weight or combine with other evidence (e.g. imagery-based cues) downstream.

4.2 Output format for Zorv's attribution model

Each candidate is emitted as a structured record — MMSI, IMO (if known), vessel name/type, the individual evidence scores (spatial, temporal, motion), the combined score, and the raw supporting values (distance, time-since-passage, heading delta, SOG at closest approach) — so the attribution layer receives ranked, scored evidence rather than a black-box single number. A simple JSON-per-candidate array is the natural interchange format.

```
[
  {
    "mmsi": "...", "imo": "...", "vessel_type": "tanker",
    "spatial_score": 0.87, "temporal_score": 0.74, "motion_score": 0.65,
    "combined_score": 0.79,
    "evidence": {
      "min_distance_nm": 3.2, "hours_since_passage": 5.1,
      "heading_delta_deg": 12, "sog_at_closest_knots": 1.4
    }
  },
  { "mmsi": "...", ... }
]
```

5. Spill Backtracking — Full Ownership

Question: given a spill detected at location X at time T, where did it most likely originate? This is the reverse problem to Section 4's forward vessel-matching, and the two are complementary — backtracking narrows down a plausible origin area/time, which then tightens the candidate-vessel search.

5.1 What the literature actually does

- Lagrangian particle tracking (forward): the standard approach (used by NOAA's GNOME model and academic oil-spill studies) represents the spill as many "particles", each independently advected by a combination of ocean-current velocity and a wind-driven "windage" term (empirically ~1–4% of wind speed, commonly ~3%), plus a small random-walk diffusion term.
- Backward/reverse drift: for source attribution, the same advection model is run in reverse — particles seeded at the observed spill polygon are pushed backward in time using negated current/wind vectors, tracing back toward a probable origin area and time.
- Bidirectional drift models (used in recent SAR+AIS oil-spill-source papers, e.g. Bohai Sea studies) run both forward (from each candidate vessel's position) and backward (from the observed spill) drift and compare where the two trajectories converge — this is the most rigorous version of exactly what this project needs, but it is a multi-day research-grade build.
- Real Indian-Ocean precedent: NOAA's GNOME model, forced with ECMWF winds and Indian Ocean current data (INDOFOS/HYCOM), has been used to hindcast an actual Indian-coast spill — the August 2011 MV Rak fuel-oil spill off Mumbai — and more recently the 2025 MSC ELSA III spill off Kochi has been analysed the same way. This confirms the approach is directly applicable to our geography, not just a generic technique.

5.2 What is realistically implementable tonight + tomorrow

A full Lagrangian particle simulation (GNOME-style, with proper current/wind fields, diffusion, and oil-weathering physics) is out of scope for this timeframe. A simplified reverse single-point (or small-cluster) drift estimate is implementable and gives a genuinely useful narrowing signal:

1. Take the spill centroid (lat, lon) and detection time T.
2. Pull average wind vector and average ocean-current vector for that location and time window from a free API (Section 5.3).
3. Compute a combined drift vector: current vector + a windage term (~3% of wind speed, in the wind direction) — this is the same first-order model GNOME itself is built on, just without the particle cloud, diffusion, or time-varying field.
4. Project this vector backward from the spill location over a chosen lookback window (e.g. 6–24 h, matching the vessel-matching lookback in Section 4) to estimate a probable origin point and, more usefully, a probable origin corridor (a buffered line, not a single pinpoint, to reflect the genuine uncertainty).
5. Feed that origin corridor back into Section 4 as a spatial prior — vessels whose trajectory passed through the estimated origin corridor near the estimated origin time get a boosted spatial/temporal score.

5.3 Data needed for the simplified version — and what's actually accessible tonight

- Wind: NOAA GFS / ECMWF forecast-and-recent-past wind fields, accessible with zero setup via the free Open-Meteo Weather Forecast / Historical Weather API (no API key, JSON, wind speed and direction at 10 m) — by far the fastest option for a hackathon.
- Ocean currents: Open-Meteo's Marine Weather API also exposes `ocean_current_velocity` and `ocean_current_direction` directly (Copernicus Marine / MeteoFrance-based), again with no API key — this is the practical choice over NASA's OSCAR dataset, which is scientifically the more standard current product but requires an Earthdata login and NetCDF handling that isn't worth the setup time tonight.
- If Open-Meteo's marine current coverage is too coarse or unavailable for a specific Arabian-Sea/Bay-of-Bengal point, NASA/JPL's OSCAR near-real-time 0.25° surface-current product (PO.DAAC, free but requires an Earthdata account) is the documented fallback — flagged here so it's in the report even if not used.

5.4 Explicitly flagged: implementable vs. report-only

Item	Status
Reverse single-vector drift estimate (current + windage, backward-projected from spill centroid)	Implementable tonight/tomorrow — coded, feeds Section 4 as a spatial prior.
Origin corridor (buffered backward line, not a single point) to reflect uncertainty	Implementable — simple buffer around the backward-projected line.
Time-varying wind/current fields over the lookback window (vs. a single averaged vector)	Implementable if time allows (stretch goal) — otherwise use one averaged vector for the whole window and say so explicitly in the demo.
Full Lagrangian particle-cloud simulation with diffusion (GNOME-style)	Report-only — describe the method and cite GNOME/the MV Rak and MSC ELSA III precedents; not coded.

Item	Status
Bidirectional forward+backward drift convergence matching (SAR+AIS literature method)	Report-only — the rigorous version; explicitly out of scope for this timeframe, documented as future work.
Oil weathering (evaporation, spreading, emulsification) affecting drift over time	Report-only — acknowledged as a real factor GNOME/ADIOS model, not attempted here.

Table 3. Backtracking scope: what gets coded vs. what is documented as a limitation/future-work item.

6. GIS Tooling

All five are standard, actively maintained, pip-installable Python libraries — chosen for what they do in this specific pipeline, not as a generic GIS-stack list.

Library	Specific use in this project
GeoPandas	Primary container for vessel trajectories and the spill polygon as GeoDataFrames — spatial joins (points-within-buffer for the spatial filter step in Section 4), clipping trajectories to the Indian EEZ / Arabian Sea / Bay of Bengal area of interest (Section 3, step 8), and exporting the ranked candidate list with geometry attached for the frontend map.
Shapely	Underlying geometry engine for the specific operations in Sections 3–5: building the spill polygon and its uncertainty buffer, constructing each vessel's trajectory as a LineString, running <code>.intersects()</code> / <code>.distance()</code> for the trajectory-intersection and distance-from-spill checks, and building the backward-drift "origin corridor" as a buffered LineString.
PyProj	Coordinate handling correctness — converting between WGS84 (lat/lon, what AIS reports) and an equal-distance/equal-area projected CRS before running any distance or buffer calculation, since haversine-on-raw-degrees is only an approximation and Shapely buffers assume a planar CRS. Used specifically before the Section 4 distance scoring and the Section 5 buffer step.
Rasterio	Only needed if the spill-detection team's satellite imagery output (SAR/optical oil-slick mask) is handed to us as a raster (GeoTIFF) rather than a pre-vectorised polygon — used to read that raster and extract/vectorise the spill footprint into the polygon Sections 4–5 need. If the imagery team already hands off a polygon/GeoJSON, Rasterio isn't needed at all — flagged as conditional, not a hard dependency.
MovingPandas	Turns the cleaned, sorted AIS points from Section 3 directly into TrajectoryCollection objects — handles the observation-gap splitting (Section 3, step 5) and trajectory smoothing/simplification out of the box, which otherwise we'd hand-roll; also gives fast trip-level analytics (start/end points, stop detection) useful for the loitering-detection part of Section 4's motion score.

Table 4. GIS libraries mapped to their specific role in this project's pipeline.

7. Relevant Papers & Projects

1. Bidirectional drift model for oil-spill source tracing (SAR + AIS)

Multiple related papers (Bohai Sea, China, 2021–2024) filter AIS data to the spill area, convert the oil-spill footprint into trajectory points, and run a Lagrangian-model-based bidirectional (forward + backward) drift comparison to identify the most likely responsible vessel among several candidates. Directly relevant to both Section 4 (matching) and Section 5 (backtracking) of this brief — this is the closest published equivalent to the full pipeline we are approximating in a simplified form. [ScienceDirect — "A new ship tracing technology from oil spills based on multi-source data"](#)

2. Real Indian-Ocean case study using GNOME (drift-model precedent for Section 5)

Prasad, S.J. et al., "Oil spill trajectory prediction with high-resolution ocean currents," *Journal of Operational Oceanography* (2019) — hindcasts the August 2011 MV Rak fuel-oil spill off Mumbai using NOAA's GNOME model forced with ECMWF winds and Indian Ocean current models (INDOFOS/HYCOM). Confirms the wind+current drift approach described in Section 5 has been validated on an actual Arabian Sea incident, not just used generically. [Taylor & Francis Online](#)

3. AIS + satellite imagery integration for oil-spill detection and accountability

"Tracing illegal oil discharges from vessels using SAR and AIS in Bohai Sea of China," *Marine Pollution Bulletin* — proposes a framework combining SAR imagery, AIS data, and marine-environment data for illegal-discharge accountability; useful precedent for how the spill-detection team's SAR output and our AIS matching layer should hand off to each other. [ScienceDirect](#)

4. AIS-based oil-spill environmental risk mapping

Zhu, G. et al., "Oil Spill Environmental Risk Assessment and Mapping in Coastal China Using AIS Data," *Sustainability* 14(10), 5837 (2022) — combines AIS traffic density with spill-accident records to map risk; useful background for weighting vessel-type priors in Section 4.2. [DOI: 10.3390/su14105837](#)

5. MovingPandas — official ship-trajectory worked example

MovingPandas' own documentation includes a ship-data analysis tutorial using Danish Maritime Authority AIS data — observation-gap splitting, trip extraction, and trajectory visualisation, i.e. almost exactly Section 3's pipeline, already implemented against the same dataset recommended as Source 3. Directly reusable as a starting point tonight. [movingpandas.github.io — Ship data analysis example](#)

8. Sources

All links below were checked directly (not from memory) while preparing this brief.

- [AISstream.io — documentation & free API](#)
- [Global Fishing Watch — API documentation](#)
- [Global Fishing Watch — data availability & download portal](#)
- [Danish Maritime Authority — AIS data download](#)
- [NOAA Office of Response & Restoration — GNOME oil-spill trajectory model](#)
- [Prasad et al. \(2019\), "Oil spill trajectory prediction with high-resolution ocean currents" \(MV Rak, Mumbai case study\)](#)
- ["A new ship tracing technology from oil spills based on multi-source data" — ScienceDirect](#)
- ["Tracing illegal oil discharges from vessels using SAR and AIS in Bohai Sea of China" — ScienceDirect](#)
- [Zhu et al. \(2022\), "Oil Spill Environmental Risk Assessment and Mapping in Coastal China Using AIS Data" — Sustainability](#)
- [Open-Meteo — free Weather Forecast & Marine Weather APIs \(wind, ocean current velocity/direction, no API key\)](#)
- [NASA PO.DAAC — OSCAR ocean surface current dataset \(fallback current source\)](#)
- [MovingPandas — official ship-trajectory analysis example \(Danish AIS data\)](#)

Summary — Final Output Checklist

Deliverable	Where in this document
AIS explanation + which fields matter	Section 1
2–3 usable AIS sources	Section 2
Trajectory reconstruction method	Section 3
Spill-to-vessel matching method (spatial/temporal/motion), scored/ranked for Zorv	Section 4
Backtracking approach — feasible vs. report-only	Section 5
Recommended GIS libraries + specific use per library	Section 6
2–3 relevant papers/projects	Section 7
Sources	Section 8